Whole-packet PPM, Hybrid A or Hybrid B? Results and a suggestion

Walter K6ATV, 28 October 2026. Aggregates only: no callsigns or positions. Code to reproduce every table: tools/hybrid/ in this repository (branch hybrid).

Summary

- **Three layouts, one table size.** Whole packet (v2.0.0), **Hybrid A** (native path block; PPM of source, destination and info) and **Hybrid B** (native source, destination and path; PPM of the info field only). Each was tested with a model trained for it, on stations the model never saw, at about 100 KB of table.
- **Size, heard directly (bytes saved vs plain):** N=2: whole packet 56.9%, A 55.0%, B 53.1%. N=4: 62.6%, 60.6%, 57.1%. Full frames only: 45.8%, 44.1%, 45.3%. B matches A on full frames and is 2 to 3.5 points behind once priming is used. B's table is smaller (79 KB against 100 to 105 KB). Moricef's draft-0 native encoder saves 27% on the same packets.
- **Why a hybrid at all:** a relay can add its RXT tuple and send the frame on without decoding PPM. Whole-packet relays took 46 ms mean (162 ms worst case) on an ESP32 and cannot forward a primed frame they have no reference for. In mixed direct and relayed reception, A delivers 5 to 12 points more of the originators' packets.
- **RXT (moricef's design):** each relay adds a 4-byte tuple and the frame one mask byte, in every layout. On every relayed frame it costs about 8 points of saving; on full frames only, about 4 points at N=2 and 2 at N=4.
- **Suggestion:** choose B if the source callsign must be readable on every frame (identification); otherwise A is 2 to 3.5 points smaller. RXT after the PPM payload, mask byte last, on full frames only. SF and total output power in a capability beacon, with the RXT tuple unchanged. Change nothing on air while the current compression test runs.

What was tested

Packets. 93,145 LoRa APRS packets from 1,669 stations (stations whose callsign hash falls in buckets 0 to 2 of 10), captured read-only from APRS-IS on 24 October. Every frame was decoded by a receiver holding only what it had been sent, and compared with the original: 0 failures. Sizes are bytes saved against the plain frame including the 3-byte radio prefix, RXT not counted unless stated. N is how often a station sends a full frame (every Nth transmission); the others are primed. The percentages are totals over all packets, so they include repeated beacons from fixed stations.

Models. The three PPM tables were trained for this comparison with the repo's trainer (with an added option for the text form) on LoRa captures from the other 70% of stations only (about 175,000 unique packets each), with the same minimum count (128), so no test station was in any training set. The text each model learns matches its layout: the whole packet text, source and destination and info (no path), or the info field with a leading colon. Table sizes (the trainer's flash estimate): whole 105 KB, A 100 KB, B 79 KB. **Why not the shipped model v1:** v1 was trained on all stations, so some test stations were in its training data. That inflates the sizes and favors layouts closest to its training text, so v1 is not used for these tables.

Credit. RXT is moricef's design: digipeaters appending per-hop reception measurements (RSSI, SNR, frequency offset, time to hop) to the packet, his RXT v1 and v2 formats, and the native-frame version with its mask byte that ties each measurement to the relay that made it (docs RXT.md and RXT_V2.md in moricef/LoRa_APRS_iGate, branch rxt). The native part and several other ideas here also come from his native LoRa APRS frame (that repository and moricef/LoRa_APRS_Tracker, branch devel) and from his

message with measurements on his Ariège and Arizona captures: a native frame that structures the fields, the RXT tuple and mask byte as he defined them, packed callsigns and short path codes, keeping the RXT block outside the compressed payload, 'source in clear, then PPM of the rest' (the model for hybrid B), and the relay timing method (decode, edit and encode per packet on a TTGO LoRa32 V2.1). The native bytes here are counted with simplified stand-ins for those fields (a 1-byte path code, a 5-byte packed callsign, a 1-byte destination code, a 6-byte block after a relay hop), not with his encoder; his encoder is used only for the 'native' column. Hybrid A, the matched-table comparison, the mixed-reception model and this harness are new in this work.

The three layouts:

	Whole-packet PPM (v2.0.0)	Hybrid A	Hybrid B
Native part (a relay reads and edits it)	Type byte only	Type byte, path block (path code, used marks, relay callsigns); RXT tuples and mask byte after the PPM payload	Packed source (5 bytes), destination code (1 byte), path block; the codec prefix carries the type; RXT as in A
PPM part	Everything: source, destination, path, info	Source, destination and info	Info field only
Who decompresses	Originator encodes; every relay decodes, edits the path and re-encodes; the iGate decodes	Originator encodes; relays copy the PPM bytes; the iGate decodes	Same as A
Priming reference	Whole text including path	Text without the path, so it is the same for every copy	The info field; the source is readable, so no station id is needed on primed frames

1. Size when a station is heard directly

Bytes saved vs plain	Native (moricef, draft 0)	Whole packet (105 KB)	Hybrid A (100 KB)	Hybrid B (79 KB)
Full frames only (N=1)	27%	45.8%	44.1%	45.3%
N=2	27%	56.9%	55.0%	53.1%
N=4	27%	62.6%	60.6%	57.1%
N=2, moving stations	32%	48.4%	44.9%	43.9%
N=2, stationary stations	26%	62.2%	60.4%	58.0%

Native column: moricef's own native encoder (tracker repo, branch devel, 20 October 2026), unchanged, on the same packets; every native frame decoded back. Native has no priming and no training, so its figure does not depend on N. His rxt branch (26 October) has the same Huffman table and text coding. His newer variant (free text coded at best, 41.5% on his Ariège captures against 47.7% for whole-packet PPM) is not in either branch, so it was not tested. Size only: the native format also carries structured fields and ties each RXT measurement to its relay.

A is about 2 points behind whole packet at every N. B is slightly ahead of A on full frames (45.3% against 44.1%), then falls behind as priming does more of the work: about 2 points at N=2 and 3.5 at N=4. A larger B table helps little (a 118 KB model gains about 0.4 point).

2. Hybrid B and the cost of a readable source

In B the native part is about 7 bytes on every frame: 5 bytes for the packed callsign and SSID, 1 byte for a destination code (14 common destinations, escape for the rest) and a 1-byte path code. The codec's own 3-byte prefix already carries the type byte, so no separate header byte is added. Primed frames drop the 1-byte station id, because the source is readable. About a third of info-only fields (34% of full frames in the

timing sample) are too short to shrink and go out uncompressed. In A and in whole packet, a primed frame carries only that 1-byte station id (a CRC-8 of the callsign, about 1 in 256 stations collide); the callsign comes out only with the earlier full frame. That is the difference that matters for identification (ARCEP 2012-1241 article 4, 47 CFR 97.119). B pays for a readable callsign on every frame with about 2 points at N=2 and 3.5 at N=4 against A, and 4 to 5.5 against whole packet. The smaller table (79 KB against 100 to 105 KB) is a side benefit of an info-only model.

3. Relayed frames and RXT

RXT (moricef's reception-measurement extension): each relay appends one 4-byte tuple (RSSI, SNR, frequency offset, TTH) and the first relay one mask byte, so a frame relayed R times grows by 4R+1 bytes. The originator adds none, because it received nothing; the final receiver reports its own measurement off air. For packets whose path starts with WIDE1-1, repeated by one relay that replaces it with its own callsign (hops simulated), bytes saved vs plain without RXT, whole packet / A / B:

	No RXT	RXT on every relayed frame	RXT on full frames only
N=2	49.6% / 46.5% / 45.1%	41.7% / 38.6% / 37.2%	45.7% / 42.5% / 39.5%
N=4	57.0% / 51.7% / 48.4%	49.1% / 43.8% / 40.5%	55.0% / 49.7% / 44.0%

Every extra relay adds 4 more bytes. The baseline is a plain frame without RXT; if the plain frame carries RXT too, the percentages would be higher. At SF12, 5 bytes is about one more LoRa block (164 ms). RXT on full frames only keeps primed frames short, at the price of one measurement per N frames per station. Whole packet is assumed to re-encode at each relay with that relay's own priming state, which favors it; a relay without the reference could not forward at all.

4. Mixed direct and relayed reception

This section compares whole packet with Hybrid A only; Hybrid B was not simulated here. The decode shares do not depend on the table; the air-byte column used the shipped model v1 and is indicative only. A receiver that hears a station directly and also through a relay. The receiver keeps one reference per source (the last full frame it decoded), as the firmware's packet store does, so a relay's full frame replaces the originator's reference and the originator's next primed frame no longer matches. Whole-packet relays also cannot forward a primed frame when they missed its full frame. In the hybrid the reference text is the same for every copy, and relays forward without decoding. Each link works independently with the stated probability, and the same random draws are used for both layouts.

Case	N	Whole packet: originator packets decoded	Hybrid: originator packets decoded	Air bytes per decoded packet (whole / hybrid)
A: heard directly half the time, relay hears 90%	2	81.6%	86.7%	59.8 / 62.7
A: heard directly half the time, relay hears 90%	4	74.1%	84.6%	57.9 / 58.3
B: weak links: 70% direct, 60% relay	2	65.0%	73.3%	59.4 / 61.3
B: weak links: 70% direct, 60% relay	4	56.7%	69.1%	58.4 / 59.1
C: mostly through the relay: 30% direct	2	60.1%	65.9%	75.3 / 77.7
C: mostly through the relay: 30% direct	4	50.5%	61.1%	77.9 / 76.1

In case A, whole-packet relays could not forward 5% (N=2) to 7% (N=4) of the packets they heard because they lacked the reference, and in case B 19% to 28%. The air bytes per decoded packet are within about 3 bytes for the two layouts, because the hybrid relay also forwards frames the whole-packet relay would have dropped, and a relayed hybrid frame is 5 bytes longer. With direct-only or relay-only reception, both layouts decode everything in this model.

5. Relay time on an ESP32

Spare ESP32-PICO-D4 (same chip as the TTGO LoRa32 V2.1) at 240 MHz, using the iGate firmware's own compression library. Each packet is decoded, the path edited and the frame re-encoded, timed together. 400 random full frames and 300 consecutive same-station pairs for primed frames, after a warm-up pass. Every decode matched the original.

	mean	95th pct	max
Full frame, whole packet	46.0 ms	108.9 ms	131.6 ms
Primed frame, whole packet	57.9 ms	139.8 ms	162.4 ms
Hybrid relay (path edit + copy)	far below 1 ms		

Moricef measured 39.6 ms (Ariège) and 61.2 ms (Arizona) mean for whole-packet PPM, so the range agrees. The hybrid figure is a stand-in (a copy of about 6 + 60 bytes), not a real frame parser, so I would say 'far below 1 ms' and not quote a number. At SF7 a frame lasts about 130 ms, so the primed worst case is longer than the frame. At SF12 (about 2 s per frame) it matters little.

Encode and decode time of the codec alone

Same ESP32-PICO-D4 at 240 MHz, one firmware build per layout with its own table, the same 400 packets and 300 same-station pairs in that layout's text form, after a warm-up pass. Milliseconds, mean with 95th percentile in brackets; all frames decoded back to the original (0 mismatches).

	Encode full	Decode full	Encode primed	Decode primed	Firmware image
Whole packet (400 / 300 packets)	21.6 (51)	21.0 (51)	27.7 (66)	27.0 (65)	471 KB
Hybrid A (400 / 300)	19.5 (51)	18.9 (50)	25.3 (65)	24.6 (65)	460 KB
Hybrid B (264 / 227 that compressed)	20.7 (47)	20.1 (46)	24.8 (59)	24.2 (59)	422 KB

Time follows the number of bytes the coder processes (about 0.27 ms per byte), not the table size, so A, which codes about 9% fewer bytes, is about 10% faster. Primed frames are slower than full ones. For B, 136 of 400 full frames and 73 of 300 primed ones did not compress (an info field alone is often too short to shrink), so B's times cover only longer packets and cannot be compared directly; coding about 28% fewer bytes should make B's average time about 28% lower, which this run does not confirm. The differences in image size show the table: B is 49 KB smaller than whole packet.

6. What I suggest

- **A or B:** B if the source callsign must be readable on every frame. It costs about 2 points at N=2 and 3.5 at N=4 against A, nothing on full frames (B is a point ahead there), and gives a smaller table (79 KB; the ESP32 image is 49 KB smaller than whole packet). A otherwise. Whole packet stays smallest when relays only forward.
- **RXT:** after the PPM payload with the mask byte last, so a relay appends without moving bytes. On full frames only, if saving matters. Keep the mask: a relay can skip adding a tuple, as moricef's RXT v2 notes say.

- **SF and power:** put the SF and the station's total output power (transmit power plus antenna gain, EIRP in dBm) in a capability beacon: a short frame a station sends on boot, on a settings change and every few hours, stating its role, compression, SF, bandwidth, coding rate and power. Leave the RXT tuple unchanged, frequency offset included. The tuple has only one free bit per byte, so adding SF there would break current decoders. A collector pairs a hop's RSSI with the previous station's advertised power for path loss and uses the relay's SF to scale TTH.
- **Type bytes:** for A, bit 5 of the PPM types (0xA1 full, 0xE1 primed), leaving 5 bits for the model version. B is really moricef's native frame (0x02) with a flag saying the info field is PPM-coded. Also reserve a frame type for a capability beacon, since 0x02 is already moricef's native frame: 01 text, 02 native, 03 capability, 80 to FF PPM.
- **Defaults:** full frame every 2nd transmission for trackers, every 4th for iGates and digipeaters.
- **Next measurements, in order:** (1) a real hybrid relay in the iGate branch, timed; (2) moricef runs A and B on his Ariège and Arizona captures; (3) mixed reception again with the firmware's 24-entry packet store and Hybrid B; (4) for B, skip PPM for info fields too short to shrink, and drop the codec's duplicate prefix and length where the native header already has them.
- **Do not change on-air formats now.** Stations in the current compression test should not change on-air formats; a new frame type is a v3 discussion.

7. Limits

- The captures have no timestamps, and each packet comes from the first iGate that delivered it, so its path is as that iGate saw it. Relay hops are simulated.
- The three PPM tables were trained on about 175,000 unique packets at one minimum count; no other sizes or settings were tuned. Savings count every beacon, so they exceed the documented 43.8% held-out figure, which uses unique packet texts. On truly unseen stations, retrained tables did no better than v1 at the same size in an earlier test (per-country notes); that was not rerun here.
- Mixed reception is a simple model: independent links, one relay, one receiver, unlimited reference storage, A against whole packet only. The hybrid relay timing is a stand-in. The 6-byte relayed path block is an assumption. B assumes the codec's 3-byte prefix doubles as the native type byte.
- The legal readings (French ARCEP 2012-1241 and 47 CFR 97.113(a)(4), 97.309(a)(4), 97.119) are interpretations and not legal advice. I fetched the eCFR text through a summarizing tool, so check the wording on ecfr.gov.